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PRINTING SYSTEM

Abstract

A printing system includes a printing device, an accommodation rack in which multiple shelf sections accommodate a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections, and which has an opening in a side surface parallel to a movement direction of the support frame, a delivery device including a feeding section capable of feeding in or out the support frame in the movement direction in a state of abutting on the support frame in the accommodation rack, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening, and an exchange control section including a position sensor configured to detect an entering state of the feeding section and detect that there is no support frame in the shelf sections.

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Background/Summary

TECHNICAL FIELD

[0001] The present description discloses a printing system.

BACKGROUND ART

[0002] Conventionally, an exchangeable printing system is known in which an exchanging member (screen mask or the like) is placed on a support frame between an accommodation device having multiple accommodation sections for accommodating a support frame and a printing device (refer to Patent Literature 1).

Patent Literature

[0003] Patent Literature 1: WO 2019/225002

BRIEF SUMMARY

Technical Problem

[0004] In such a printing system, it is desirable to feed out the support frame to the printing device after confirming that there is the support frame in the accommodation section, or to feed in the support frame from the printing device after confirming that there is an empty space in the accommodation section. However, in the printing system described in Patent Literature 1, it is not discussed to detect the presence or absence of the support frame in the accommodation section.

Therefore, there is a concern that the support frame may be fed out to the printing device in a state where there is no support frame in the accommodation section, or the support frame may be fed in from the printing device in a state where there is no empty space in the accommodation section.

[0005] A main object of the present disclosure is to more reliably feed out or in a support frame between an accommodation rack and a printing device.

Solution to Problem

[0006] A first printing system of the present disclosure includes: [0007] a printing device; [0008] an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; [0009] a delivery device including a feeding section capable of feeding in or out the support frame in the movement direction in a state of abutting on the support frame in the accommodation rack, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening; and [0010] a first detection section including a first sensor configured to detect an entering state of the feeding section with respect to the accommodation rack, and capable of detecting that there is no support frame in the shelf sections based on a detection signal of the first sensor.

[0011] In the first printing system, it is possible to detect that there is no support frame in the shelf section based on the detection signal of the first sensor. Therefore, it is possible to prevent the support frame from being fed out to the printing device in a state where there is no support frame in the shelf section, or to prevent the support frame from being fed in from the printing device in a state where the support frame is present in the shelf section. As a result, the support frame can be more reliably fed out or in between the accommodation rack and the printing device.

[0012] A second printing system of the present disclosure includes: [0013] a printing device;

[0014] an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces

the printing device; and [0015] a lifting and lowering device including a pair of rack support members capable of supporting a lower surface of the accommodation rack from both sides in an orthogonal direction orthogonal to a movement direction of the support frame and an up-down direction in the state where the accommodation rack faces the printing device, and configured to lift and lower the rack support members, [0016] in which the support frame includes multiple holding sections arranged in the movement direction, the holding sections being capable of holding exchanging members of different sizes, and [0017] the accommodation rack includes a weight provided such that a center of gravity of the accommodation rack is positioned closer to the printing device than a position supported by the rack support members in the state where the accommodation rack faces the printing device.

[0018] In the second printing system, the weight is provided in the accommodation rack such that the center of gravity of the accommodation rack is positioned closer to the printing device than the position where the accommodation rack is supported by the rack support member. Accordingly, when the accommodation rack is lifted and lowered together with the rack support member, the accommodation rack is less likely to be inclined to the side opposite to the printing device. It is particularly effective in the case where an exchanging member larger in size than the exchanging member held by the holding section on the downstream side in the movement direction of the support frame is held in the holding section on the upstream side in the movement direction of the support frame. This is because the center of gravity of the accommodation rack is likely to be biased to the side opposite to the printing device with respect to the rack support member as compared with a case where the exchanging member having the same size is held in the holding sections on the downstream side and the upstream side or a case where the exchanging member is held only in the holding section on the downstream side in the movement direction.

[0019] A third printing system of the present disclosure includes: [0020] a printing device; [0021] an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; and [0022] a delivery device including a feeding section including one or more driving rolls made of urethane, and capable of feeding in or out the support frame in the movement direction by the driving rolls being driven in a state where the driving rolls abut on the support frame in the accommodation rack, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening.

[0023] In the third printing system, the driving roll made of urethane is driven in a state of abutting on the support frame in the accommodation rack, and thus, the support frame is fed out or in. As compared with the case where the support frame is fed out or in by a friction force of a rubber belt, abrasion powder is less likely to enter the printing device. This is because the urethane roll is less likely to wear than the rubber belt or the like.

[0024] A fourth printing system of the present disclosure includes: [0025] a printing device; [0026] an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; and [0027] a delivery device including a feeding section capable of feeding in or out the support frame in the movement direction, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening, [0028] in which the feeding section includes one or more driving rolls driven in a state of abutting on the support frame in the accommodation rack, and a belt provided at a position not in contact with the support frame to transmit power to the multiple driving rolls, and [0029] the multiple driving rolls have a lower abrasiveness than the belt.

[0030] In the fourth printing system, the support frame is fed out or in by driving the driving roll in a state of abutting on the support frame. The belt does not contact the support frame. Furthermore, the driving roll is lower in abrasiveness than the belt. Therefore, compared to the case where the support frame is fed out or in by the friction force of the belt, the abrasion powder is less likely to enter the printing device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- [0031] FIG. 1 is a schematic view of mounting system 1 including printing system 2.
- [0032] FIG. 2 is a perspective view of printing system 2.
- [0033] FIG. 3A is a perspective view of support frame 100.
- [0034] FIG. 3B is a perspective view of support frame 100.
- [0035] FIG. 4 is a schematic configuration diagram of printing system 2.
- [0036] FIG. 5 is a perspective view of support frame detection sensor 24.
- [0037] FIG. 6A is a diagram illustrating an operation of support frame detection sensor 24.
- [0038] FIG. 6B is a diagram illustrating the operation of support frame detection sensor 24.
- [0039] FIG. 7A is a perspective view of accommodation rack 40.
- [0040] FIG. 7B is a perspective view of accommodation rack 40.
- [0041] FIG. 8 is a cross-sectional view of accommodation rack 40 and printing device 10.
- [0042] FIG. 9 is a perspective view of accommodation rack 40.
- [0043] FIG. 10 is a side view of printing system 2.
- [0044] FIG. 11 is a perspective view of lifting and lowering device 52 and front-rear movement device 55.
- [0045] FIG. 12 is a longitudinal sectional view of clamp section 71.
- [0046] FIG. 13A is a longitudinal sectional view of stopper section 75.
- [0047] FIG. 13B is a cross-sectional view of stopper section 75.
- [0048] FIG. 14 is a perspective view of delivery device 90.
- [0049] FIG. 15A is a diagram illustrating an operation of position sensor 120.
- [0050] FIG. 15B is a diagram illustrating the operation of position sensor 120.
- [0051] FIG. 15C is a diagram illustrating the operation of position sensor 120.
- [0052] FIG. 16 is a block diagram illustrating an electrical connection relationship of printing system 2.
- [0053] FIG. 17 is a diagram illustrating a state in which worker W moves accommodation rack 40 to exchange unit 30.
- [0054] FIG. 18A is a flowchart illustrating an example of a member exchanging process routine.
- [0055] FIG. 18B is a flowchart illustrating an example of a support frame feeding-in process subroutine.
- [0056] FIG. 18C is a flowchart illustrating an example of a support frame feeding-out process subroutine.
- [0057] FIG. 19A is a diagram illustrating an operation of front-rear movement device 55 when accommodation rack 40 moves to an exchange position.
- [0058] FIG. 19B is a diagram illustrating the operation of front-rear movement device 55 when accommodation rack 40 moves to the exchange position.
- [0059] FIG. 19C is a diagram illustrating the operation of front-rear movement device 55 when accommodation rack 40 moves to the exchange position.
- [0060] FIG. 19D is a diagram illustrating the operation of front-rear movement device 55 when accommodation rack 40 moves to the exchange position.
- [0061] FIG. 20 is a perspective view of printing system 2 when accommodation rack 40 is at an

accommodation position.

[0062] FIG. **21** is a perspective view of printing system **2** when accommodation rack **40** is at an operation position.

[0063] FIG. **22** is a perspective view of printing system **2** in which accommodation rack **40** is lifted.

[0064] FIG. **23** is a perspective view of printing system **2** when accommodation rack **40** is at the exchange position.

[0065] FIG. **24** is a perspective view of delivery device **290**.

[0066] FIG. **25** is a side view of delivery device **290**.

DESCRIPTION OF EMBODIMENTS

[0067] The present embodiment will be described below with reference to drawings. FIG. **1** is a schematic view of mounting system **1** including printing system **2**. FIG. **2** is a perspective view of printing system **2**. FIGS. **3A** and **3B** are perspective views of support frame **100**. FIG. **4** is a schematic configuration diagram of printing system **2**. FIG. **5** is a perspective view of support frame detection sensor **24**. FIGS. **6A** and **6B** are diagrams illustrating an operation of support frame detection sensor **24**. FIGS. **7A** and **7B** are perspective views of accommodation rack **40**. FIG. **8** is a cross-sectional view of accommodation rack **40** and printing device **10**. FIG. **9** is a perspective view of accommodation rack **40**. FIG. **10** is a side view of printing system **2**. FIG. **11** is a perspective view of lifting and lowering device **52** and front-rear movement device **55**. FIG. **12** is a longitudinal sectional view of clamp section **71**. FIGS. **13A** and **13B** are longitudinal sectional and cross-sectional views of stopper section **75**. FIG. **14** is a perspective view of delivery device **90**. FIGS. **15A** to **15C** are diagrams illustrating the operation of position sensor **120**. FIG. **16** is a block diagram illustrating an electrical connection relationship of printing system **2**. In the present embodiment, a left-right direction (X axis), a front-rear direction (Y axis), and an up-down direction (Z axis) are as illustrated in the respective drawings.

[0068] Mounting system **1** includes printing system **2**, print inspection device **3**, mounting device **4**, mounting inspection device **5**, reflow device (not illustrated), and management computer **130**. Mounting system **1** is configured as a mounting line in which multiple mounting devices **4** for mounting a component on board **S** are arranged downstream of printing system **2**.

[0069] Print inspection device **3** is a device that inspects the state of viscous fluid such as solder paste which is printed on board **S** by printing device **10**. Mounting device **4** is a device that mounts a component on board **S** on which the viscous fluid is printed by printing device **10**. Mounting inspection device **5** is a device that inspects, for example, the state of the component mounted on board **S**. The reflow device is a device that heats and melts the printed viscous fluid to make the viscous fluid secure onto board **S**. Management computer **130** is a device that manages information of each device of mounting system **1**. Management computer **130** manages, for example, the progress status of each device of the production line which performs mounting process.

[0070] Printing system **2** includes printing device **10**, support frame detection sensor **24**, accommodation rack **40**, exchange unit **30**, and position sensor **120**. As illustrated in FIG. **1**, printing device **10** is a device that pushes a solder on screen mask **M** into a pattern hole formed in screen mask **M** using squeegees **14** to thereby apply (print) solder, as viscous fluid, onto board **S**, which is a processing target, below screen mask **M** through the pattern hole. Here, as the “processing target”, there is raised, for example, board **S** on which a component is mounted, a base material which is a three-dimensional object, or the like. As the “viscous fluid”, there is raised a solder paste or a conductive paste, an adhesive, or the like. Here, the following description will be made using board **S** and solder paste as examples.

[0071] As illustrated in FIG. **4**, printing device **10** includes printing section **11**, mask work section **15**, board processing section **17**, cleaning section **20**, supply section **21**, support frame detection sensor **24**, rack detection sensor **25**, operation panel **26** (refer to FIG. **1**), and printing control section **28** (refer to FIG. **16**).

[0072] Printing section **11** is a unit that is disposed at an upper stage of printing device **10** and executes a printing process to print viscous fluid on board **S** using screen mask **M**. Printing section **11** includes, as illustrated in FIG. **4**, printing head **12**, printing moving section **13** (refer to FIG. **16**), a squeegee lifting and lowering section (not illustrated), and squeegees **14**. Printing moving section **13** is configured to move printing head **12** in a predetermined printing direction (here, in the front-rear direction) and includes a guide formed in the front-rear direction, a slider configured to move along the guide, and a motor for driving the slider. Squeegee **14** is provided on a lower surface side of printing head **12** and is lifted and lowered by the squeegee lifting and lowering section. Printing section **11** has two squeegees **14** which are individually used in the front-rear direction.

[0073] As illustrated in FIG. **4**, mask work section **15** is provided between printing section **11** and board processing section **17** in the up-down direction, constituting a unit for fixedly holding screen mask **M**. As illustrated in FIG. **4**, mask work section **15** includes exchange section **27** and a mask fixing section (not illustrated). Exchange section **27** is a conveyance rod provided on printing head **12** and is lowered to a position where exchange section **27** is brought into abutment with screen mask **M** by an air cylinder (not illustrated), and an abutting section thereof configured to be brought into abutment with support frame **100** moves also in the front-rear direction as a result of being rotated. Printing head **12** and exchange section **27** are configured to be brought into engagement with support frame **100** between shelf section **41a** at a predetermined height in accommodation rack **40** and printing section **11** to move support frame **100**. The mask fixing section positions screen mask **M**, and supports and fixes screen mask **M** in a horizontal posture. Screen mask **M** is pushed to move along the guide of mask work section **15** in the front-rear direction by exchange section **27** and is fixed by the mask fixing section.

[0074] As illustrated in FIG. **4**, two board processing sections **17** are provided below mask work section **15**. Board processing section **17** is a unit that loads board **S**, positions and supports loaded board **S**, and brings board **S** into contact with and separates from screen mask **M**. Board processing section **17** includes board conveyance section **18** that conveys board **S** in the left-right direction, board support member **19** that supports board **S** from below; and a board lifting and lowering section that lifts and lowers entire board processing section **17** and board support member **19**.

[0075] As illustrated in FIG. **4**, cleaning section **20** is a unit that is provided between mask work section **15** and board processing section **17** in the up-down direction and performs a cleaning process of cleaning a rear surface of screen mask **M**. Cleaning section **20** has cleaning member **20a**, and screen mask **M** is cleaned as a result of cleaning member **20a**

[0076] being brought into abutment therewith.

[0077] Supply section **21** is a unit for supplying a solder accommodated in cartridge **23** on screen mask **M**. As illustrated in FIG. **4**, supply section **21** is provided in front of the printing head **12**. Supply section **21** applies a pressure to cartridge **23** to discharge the solder from cartridge **23**. Cleaning member **20a** and cartridge **23** are consumables and are exchanged as required.

[0078] Rack detection sensor **25** constitutes a sensor for detecting whether accommodation rack **40** exists in a front interior part of printing device **10**. As illustrated in FIG. **4**, rack detection sensor **25** is provided at a lower part of the front surface of printing device **10**. When weight **49** of accommodation rack **40** is in contact with rack detection sensor **25**, rack detection sensor **25** outputs an ON signal to printing control section **28** (refer to FIG. **16**). Meanwhile, when weight **49** of accommodation rack **40** is not in contact with rack detection sensor **25**, rack detection sensor **25** outputs an OFF signal to printing control section **28**.

[0079] Operation panel **26** is a unit that receives an input from worker **W** and presents information to worker **W**. Operation panel **26** includes a display section, which is a display, and an operation section of a touch panel type having a button.

[0080] Printing control section **28** is configured as a microprocessor which is made up mainly of CPU and controls entire printing device **10**. As illustrated in FIG. **16**, printing control section **28** outputs a signal to management computer **130** as an external device in addition to printing section

11, mask work section **15**, board processing section **17**, cleaning section **20**, supply section **21**, operation panel **26**, and exchange unit **30**. In addition to printing section **11**, mask work section **15**, board processing section **17**, cleaning section **20**, supply section **21**, support frame detection sensor **24**, rack detection sensor **25**, operation panel **26**, and exchange unit **30**, printing control section **28** inputs a signal from management computer **130** as an external device. Printing control section **28** has a storage section that stores information on board **S** on which a printing process is executed, a printing process program for executing a printing process on board **S**, a member exchanging process program for causing exchange section **27** to execute an exchange process of support frame **100**, and the like.

[0081] As illustrated in FIGS. **5**, **6A**, and **6B**, support frame detection sensor **24** is an optical sensor capable of detecting an object (for example, support frame **100**) positioned (hereinafter referred to as a movement path) between accommodation rack **40** and printing device **10**. Support frame detection sensor **24** includes light projecting section **24a** provided on the left side of the movement path and light receiving section **24b** (refer to FIGS. **6A** and **6B**) provided on the right side of the movement path. When there is an object (for example, support frame **100**) on the movement path, the object blocks light from light projecting section **24a** and support frame detection sensor **24** outputs an OFF signal to printing control section **28** (refer to FIG. **16**). Meanwhile, when there is no object in the movement path, the light from light projecting section **24a** is received, and the ON signal is output to printing control section **28**.

[0082] Accommodation rack **40** is an accommodation body that accommodates support frame **100** capable of supporting exchanging members (for example, screen mask **M** and board support member **19**) related to the printing process performed by printing section **11**. Accommodation rack **40** has, as illustrated in FIGS. **7A**, **7B**, and **8**, rack main body **41**, handle **42**, caster section **43**, supporting and fixing member **44**, and weight **49**.

[0083] As illustrated in FIG. **3A**, accommodation rack **40** can accommodate support frame **100** to which screen mask **M** is fixed. In the present embodiment, in screen mask **M**, a pattern hole formed in front of support frame **100** is larger in size than the pattern hole formed in the rear of support frame **100**. As illustrated in FIG. **3B**, accommodation rack **40** can accommodate support frame **100** that holds board support member **19**. Board support member **19** is held by holding section **101** provided in support frame **100**. Holding section **101** includes pair of claw members **103** provided so that tip portions of claw members face each other. Pair of claw members **103** supports the lower surface of board support member **19**. Pair of claw members **103** is normally biased upward by a spring (not illustrated), and is in a horizontally maintained closed state. When pair of claw members **103** is pushed down from above, pair of claw members **103** is in an open state of being rotated downward against the elastic force of the elastic body. Each holding section **101** can hold board support member **19** of a different size. For example, in the present embodiment, board support member **19** held by holding section **101** provided at the front side of support frame **100** is larger in size than board support member **19** held by holding section **101** provided at the rear side of support frame **100**.

[0084] Rack main body **41** includes multiple (four in the present embodiment) shelf sections **41a**, passage section **41b**, and multiple opening portions **41c**. Shelf sections **41a** are arranged at predetermined intervals in the up-down direction so that support frame **100** can be placed in the up-down direction. Shelf sections **41a** are provided on the left and right inside rack main body **41** so that end portions of support frame **100** are inserted to be supported by shelf sections **41a**. Passage section **41b** communicates with the outside such as printing section **11**. Support frame **100** is movable in the front-rear direction (movement direction) through passage section **41b**, and is fed into or out of rack main body **41**. Opening portion **41c** is used to cause feeding section **91** of delivery device **90** (refer to FIG. **14**) to enter or exist from accommodation rack **40**. Opening portion **41c** is a groove formed on the left side surface of rack main body **41** and extending in the front-rear direction.

[0085] Handle **42** is made up of a rod-shaped member gripped by worker W and a support member that fixes the rod-shaped member. Caster section **43** is provided below rack main body **41**. Caster section **43** includes a structure having a space in the center thereof and casters which are provided on a lower surface side of the structure and which each have a wheel. Worker W grips this handle **42** to move accommodation rack **40** (refer to FIG. 17).

[0086] Supporting and fixing member **44** is a plate-shaped member which is fixed to front left and right end portions on a lower surface side of rack main body **41**. Supporting and fixing member **44** is a member which is used for connection to exchange unit **30**. As illustrated in FIG. 9, supporting and fixing member **44** has block section **45**, receiving section **46**, locking hole **47**, and insertion hole **48**. Block section **45** constitutes a member to which hook section **56** provided in exchange unit **30** is connected, and a space into which hook section **56** is inserted is defined between two blocks. Receiving section **46** is a portion that receives cam follower **53a** provided in rack support member **53** of exchange unit **30**, and has an arc-shaped cut portion. Locking hole **47** is a through hole into which clamp lever **72** of fixing section **70** provided in rack support member **53** of exchange unit **30** is inserted. Insertion hole **48** is a through hole into which insertion pin **76** of fixing section **70** provided in rack support member **53** of exchange unit **30** is inserted.

[0087] Weight **49** is used to position the center of gravity of accommodation rack **40** on printing device **10** side (rear side) with respect to rack support member **53** of exchange unit **30** in a state where accommodation rack **40** faces printing device **10**. As illustrated in FIGS. 8 to 10, weight **49** is provided behind the lower surface of rack main body **41**. The mass of weight **49** is determined in consideration of the weight of the entire accommodation rack **40**, the position of board support member **19**, the weight of support frame **100**, the weight of board support member **19** held by support frame **100**, and the like.

[0088] Exchange unit **30** exchanges support frame **100** between accommodation rack **40** and printing device **10** when accommodation rack **40** is lifted and lowered. As illustrated in FIGS. 1 and 2, exchange unit **30** includes left unit **51**, right unit **61**, and exchange control section **110** (refer to FIG. 16). Left unit **51** is provided at a front left side of a housing of printing device **10**. Right unit **61** is provided at a front right side of the housing of the printing device **10**. As illustrated in FIG. 4, these left unit **51** and right unit **61** are formed such that length A along a movement direction of support frame **100** is shorter than length B of accommodation rack **40**. Exchange unit **30** has lifting and lowering space **58** defined in the center between left unit **51** and right unit **61** for lifting and lowering accommodation rack **40**. As illustrated in FIG. 4, exchange unit **30** accommodates accommodation rack **40** on the same surface as a front surface of a main body of exchange unit **30** or further inwards than the front surface of the main body of exchange unit **30** when accommodation rack **40** stays in an accommodation position, whereas when accommodation rack **40** stays in an exchange position, exchange unit **30** supports accommodation rack **40** in a position which protrudes further than the front surface of the main body of exchange unit **30**.

[0089] As illustrated in FIG. 11, left unit **51** includes lifting and lowering device **52**, front-rear movement device **55**, fixing section **70**, delivery device **90** (refer to FIG. 16), and exchange control section **110** (refer to FIG. 16).

[0090] Lifting and lowering device **52** is a device that lifts and lowers accommodation rack **40** between an exchange position when support frame **100** is exchanged between accommodation rack **40** and printing device **10** and an operation position below the exchange position. As illustrated in FIG. 11, lifting and lowering device **52** includes rack support member **53** and lifting and lowering drive section **54**. Rack support member **53** is a plate-shaped member that receives accommodation rack **40** and supports accommodation rack **40** from the left side of the lower surface. Rack support member **53** is provided on a surface of left unit **51** facing right unit **61**. Rack support member **53** is provided with cam follower **53a** at front and rear end portions thereof. Cam follower **53a** supports the supporting and fixing member **44** provided in accommodation rack **40** from below, and vertically moves accommodation rack **40** in accordance with the vertical movement of rack support

member 53. For example, lifting and lowering drive section 54 may be configured to have a ball screw having a lifting and lowering shaft and a drive motor to vertically move rack support member 53 or may be configured to vertically move rack support member 53 using a linear motor. [0091] Front-rear movement device 55 is configured to move accommodation rack 40 between the operation position where accommodation rack 40 can be lifted and lowered by lifting and lowering device 52 and the accommodation position where accommodation rack 40 enters the interior part of the housing of printing device 10. Front-rear movement device 55 moves accommodation rack 40 in the horizontal direction. As illustrated in FIG. 11, front-rear movement device 55 includes hook section 56 and front-rear drive section 57. Hook section 56 is provided to be lifted and lowered on the axis of front-rear drive section 57. Hook section 56 is a member that is lifted toward block section 45 provided in accommodation rack 40 and engages with block section 45. Front-rear drive section 57 is configured to move hook section 56 between the accommodation position and the operation position. Front-rear drive section 57 may be configured to have a ball screw and a drive motor to move hook section 56 forward and backward or may be configured to move hook section 56 forward and backward using a linear motor.

[0092] As illustrated in FIG. 11, fixing section 70 is a safety mechanism that prevents unintended movement and fall of accommodation rack 40 when accommodation rack 40 is lifted and lowered by lifting and lowering device 52. Fixing section 70 includes clamp section 71 and stopper section 75.

[0093] As illustrated in FIG. 12, clamp section 71 has clamp lever 72 configured to be inserted into locking hole 47 formed in supporting and fixing member 44 of accommodation rack 40, thereby constituting a mechanism that restricts accommodation rack 40 from moving in the horizontal direction. Clamp section 71 is provided in rack support member 53 and lifted and lowered together with rack support member 53. Clamp section 71 includes clamp lever 72, lever rotating section 73, and air cylinder 74. Clamp lever 72 has a lever configured to be inserted into locking hole 47, a rotation shaft fixed to rack support member 53 to rotate the lever, and an elliptic hole into which a pin of lever rotating section 73 is inserted. Lever rotating section 73 is a member which is coupled directly to a rod of air cylinder 74 for linear movement. A pin inserted into the elliptic hole is provided at the tip of the lever rotating section 73. Air cylinder 74 constitutes a drive section in which a rod is vertically moved by means of compressed air which is supplied from an air supply device (not illustrated). In clamp section 71, when the rod of air cylinder 74 is lifted up, clamp lever 72 rotates about the rotation shaft, whereby the lever is inserted into locking hole 47. In addition, in this clamp section 71, when the rod of air cylinder 74 is lowered, clamp lever 72 rotates about the rotation shaft, whereby the lever inserted in locking hole 47 is released therefrom.

[0094] As illustrated in FIGS. 13A and 13B, stopper section 75 has insertion pin 76 configured to be inserted into insertion hole 48 formed in supporting and fixing member 44 of accommodation rack 40, constituting a mechanism that prevents accommodation rack 40 from falling as a result of insertion pin 76 being inserted into insertion hole 48 when accommodation rack 40 is lifted. Stopper section 75 has insertion pin 76 and pin drive section 77. Pin drive section 77 causes insertion pin 76 to be inserted into insertion hole 48 when accommodation rack 40 is lifted from the operation position, whereas when accommodation rack 40 stays in the operation position, pin drive section 77 causes insertion pin 76 to be pulled in to release the insertion of insertion pin 76 into insertion hole 48. This pin drive section 77 includes spring 78, cam follower 79, guide member 80, and air cylinder 81. Insertion pin 76, spring 78, and cam follower 79 are provided in rack support member 53 and lifted and lowered together with rack support member 53. Meanwhile, guide member 80 and air cylinder 81 are provided at the operation position on the housing side of exchange unit 30, and are not lifted and lowered. Spring 78 is a member that biases insertion pin 76 towards insertion hole 48 side at all times. Insertion pin 76 constitutes a member configured to be inserted into insertion hole 48 and is a cylindrical member having an interior space. Spring 78 is provided in the internal space of insertion pin 76. Cam follower 79 is provided at the rear end of

insertion pin **76**. Cam follower **79** is brought into abutment with a wall section of guide member **80** to be lifted and lowered together with rack support member **53** while being guided by guide member **80**. Guide member **80** constitutes a member having a shape in which two members, which each have an L-shaped section and whose longitudinal direction follows the up-down direction, face each other. In guide member **80**, the L-shaped wall section is formed in the up-down direction, so that cam follower **79** moves along this wall section. Guide member **80** is fixed to a rod of air cylinder **81**. Air cylinder **81** constitutes a drive section that drives the rod in a horizontal direction by means of compressed air supplied from an air supply device (not illustrated). When the rod of air cylinder **81** is pushed out when rack support member **53** is at the operation position, stopper section **75** is in a state where insertion pin **76** biased by spring **78** protrudes from rack support member **53** (refer to FIG. **13A**). In a case where the rod of air cylinder **81** is pulled when rack support member **53** is at the operation position, stopper section **75** is brought into a state in which insertion pin **76** is pulled through guide member **80** and cam follower **79** and does not protrude from rack support member **53** (refer to FIG. **13B**). Further, stopper section **75** is such that when rack support member **53** is lifted from the operation position, cam follower **79** is released from the abutment with guide member **80**, causing a state in which insertion pin **76** is inserted in insertion hole **48** at all times by means of the biasing force of spring **78**.

[0095] Delivery device **90** exchanges support frame **100** between accommodation rack **40** and printing device **10** by driving feeding section **91** in a state in which feeding section **91** abuts on support frame **100**. As illustrated in FIG. **14**, delivery device **90** includes feeding section **91** and air cylinder **92**. Feeding section **91** includes belt **93** and drive section **94** that rotates belt **93**. Belt **93** moves support frame **100** to printing device **10** side or accommodation rack **40** side by the friction force of surface **93a**. Drive section **94** is configured as, for example, a servo motor. As illustrated in FIG. **14**, feeding section **91** is configured such that the power of drive section **94** is transmitted to belt **93** via multiple pulleys **95**. Air cylinder **92** causes feeding section **91** to enter and exit from accommodation rack **40** via opening portion **41c**. A magnet is provided on the piston of air cylinder **92**.

[0096] Moreover, right unit **61** has the same configuration as left unit **51** except that right unit **61** faces left unit **51** and does not include a delivery device, and a detailed description thereof is omitted. Right unit **61** includes a rack support member, a cam follower, a lifting and lowering drive section, a forward/backward moving section, a hook section, and a front-rear drive section.

[0097] Exchange control section **110** is configured as a microprocessor mainly including CPU, and controls entire exchange unit **30**. As illustrated in FIG. **16**, exchange control section **110** outputs signals to lifting and lowering drive section **54**, front-rear drive section **57**, and printing control section **28**. Exchange control section **110** receives signals from printing control section **28**, management computer **130**, position sensor **120**, and the like.

[0098] Position sensor **120** is a sensor capable of detecting the position of feeding section **91** in the left-right direction. Position sensor **120** includes two automatic switches **120a** and **120b**. Automatic switches **120a** and **120b** are attached to the cylinder tube of air cylinder **92**. Automatic switches **120a** and **120b** have magnetic detection elements capable of detecting magnetism. Automatic switches **120a** and **120b** detect magnetism when the magnet provided in the piston of air cylinder **92** approaches the magnetoresistive element. As illustrated in FIG. **15A**, in a state in which the piston of air cylinder **92** is positioned at the leftmost end (hereinafter referred to as a “backward end”), automatic switch **120a** outputs an ON signal to exchange control section **110**, and automatic switch **120b** outputs an OFF signal to exchange control section **110**. At this time, feeding section **91** is positioned outside accommodation rack **40**. As illustrated in FIG. **15B**, when the piston of air cylinder **92** is moved from the backward end side to the right side in a state in which support frame **100** is not accommodated in shelf section **41a**, feeding section **91** enters the inside of accommodation rack **40** (hereinafter, this state is referred to as “swing”), and the piston is positioned at the rightmost end (hereinafter referred to as a forward end). At this time, automatic

switch **120a** outputs an OFF signal to exchange control section **110**, and automatic switch **120b** outputs an ON signal to exchange control section **110**. As illustrated in FIG. **15C**, when the piston of air cylinder **92** is moved from the backward end side to the forward end side in a state where there is support frame **100** in shelf section **41a**, feeding section **91** abuts on support frame **100**, and the piston is positioned between the backward end and the forward end. At this time, automatic switches **120a** and **120b** output an OFF signal to exchange control section **110**.

[0099] Next, a printing process of printing device **10** configured as described heretofore will be described with reference to FIG. **17**. Worker W performs work of mounting accommodation rack **40** in exchange unit **30** to execute a printing process. FIG. **17** is a diagram illustrating a way by which worker W moves accommodation rack **40** to exchange unit **30**. Worker W inputs an execution of a printing process from operation panel **26** after worker W has mounted accommodation rack **40** in exchange unit **30**. A printing process routine for executing the printing process is stored in the storage section of printing control section **28**, and the printing process routine is executed after printing control section **28** has received a printing process execution command as an input from worker W. When the printing process routine is started, firstly, CPU of printing control section **28** executes a process of bringing board S into abutment with screen mask M by causing board processing section **17** to convey, fix, and lift board S. For this, CPU causes mask work section **15** to adjust the position of screen mask M to thereby perform a position alignment between a pattern hole and board S. Subsequently, the CPU moves cartridge **23** to above screen mask M and then causes cartridge **23** to discharge the solder. Then, the CPU causes squeegees **14** to be lowered while causing printing head **12** to move, so that squeegees **14** are brought into abutment with an upper surface of screen mask M while being caused to move in the front-rear direction to print the solder on board S. In this way, printing control section **28** performs the printing process including the board conveyance and fixation process, the solder supply process, and the squeegee moving process to thereby execute printing on board S. The CPU repeatedly executes this printing process while exchanging screen mask M until the production process of producing all types of boards S in the production plan is completed.

[0100] Next, the operation of exchange unit **30** for exchanging support frame **100** will be described with reference to FIGS. **18** to **23**. FIG. **18A** is a flowchart illustrating an example of a support frame exchanging process routine. FIG. **18B** is a flowchart illustrating an example of a support frame feeding-in process subroutine. FIG. **18C** is a flowchart illustrating an example of a support frame feeding-out process subroutine. FIGS. **19A** to **19D** are diagrams illustrating the operation of front-rear movement device **55** when accommodation rack **40** moves to the exchange position. FIG. **20** is a perspective view of printing system **2** when accommodation rack **40** is in the accommodation position. FIG. **21** is a perspective view of printing system **2** when accommodation rack **40** is at the operation position. FIG. **22** is a perspective view of printing system **2** in which accommodation rack **40** is lifted. FIG. **23** is a perspective view of printing system **2** when accommodation rack **40** is at the exchange position.

[0101] The exchange of the member is performed by collecting screen mask M, collecting board support member **19**, supplying new board support member **19**, and supplying new screen mask M. The operation of collection and supply is performed using a common movement path. The member exchanging process routine is stored in a storage section of exchange control section **110** and is executed after exchange unit **30** has been activated. When this routine is executed, CPU of exchange control section **110** determines based on a detection signal from rack detection sensor **25** whether accommodation rack **40** is disposed at the accommodation position (S100). As illustrated in FIG. **20**, when accommodation rack **40** is disposed at the accommodation position, the CPU lifts hook section **56** and connects hook section **56** to block section **45** as illustrated in FIGS. **19A** and **19B** (S110). When hook section **56** is inserted into block section **45**, front-rear movement device **55** can move accommodation rack **40** forwards and backwards. Meanwhile, when it is determined in S100 that accommodation rack **40** is not disposed at the accommodation position, the CPU waits

until accommodation rack **40** is disposed at the accommodation position.

[0102] After **S110**, the CPU of exchange control section **110** waits until a timing for executing the exchange of screen mask **M** as the exchangeable member is reached (**S120**). Examples of the timing for exchanging screen masks **M** include a timing after the production of boards **S** of a predetermined type ends, and a timing before the production of boards **S** of a subsequent type starts.

[0103] Meanwhile, when an affirmative determination is made in **S120**, the CPU moves hook section **56** to the operation position by front-rear movement device **55** (**S130**), as illustrated in FIG. **19C**. Then, as illustrated in FIG. **21**, accommodation rack **40** is pulled out to be lifted.

[0104] Next, as illustrated in FIG. **19D**, the CPU lifts rack support member **53** and connects rack support member **53** to the supporting and fixing member **44** of accommodation rack **40** (**S140**). When rack support member **53** is lifted at the operation position, cam follower **53a** enters receiving section **46**, and accommodation rack **40** is connected to rack support member **53**.

[0105] Next, the CPU executes a clamp process (**S150**). Specifically, the CPU drives air cylinder **81** so that insertion pin **76** of stopper section **75** is inserted into insertion hole **48** when air cylinder **74** is driven so that clamp lever **72** is inserted into locking hole **47**.

[0106] In the present embodiment, holding section **101** arranged at the front side of support frame **100** holds board support member **19** having a larger size (heavier) than holding section **101** arranged at the rear side of support frame **100**. Therefore, the center of gravity of accommodation rack **40** tends to be biased toward the front side. However, in the present embodiment, accommodation rack **40** has weight **49** behind the lower surface. Therefore, even after rack support member **53** is lifted and cam follower **53a** enters receiving section **46**, the center of gravity of accommodation rack **40** is closer to printing device **10** side than the front cam follower **53a**, and accommodation rack **40** is less likely to be inclined forward with respect to rack support member **53**. As a result, the position of locking hole **47** to clamp lever **72** is less likely to shift, and the position of insertion hole **48** to insertion pin **76** is less likely to shift. Thus, even in such a case, the clamp process can be executed relatively easily.

[0107] Subsequently, as illustrated in FIGS. **22** and **23**, the CPU of exchange control section **110** controls lifting and lowering drive section **54** such that empty shelf section **41a** faces exchange door **29** (**S160**). Next, the CPU executes the support frame feeding-in process subroutine (refer to **S170** and FIG. **18B**). When the support frame feeding-in process subroutine is started, the CPU outputs a feeding-out instruction of support frame **100** to printing device **10** (**S300**). When the feeding-out instruction of support frame **100** is input, printing control section **28** executes a process of moving screen mask **M** to empty shelf section **41a** by using exchange section **27**.

[0108] Subsequently, the CPU of exchange control section **110** controls air cylinder **92** such that feeding section **91** of delivery device **90** enters accommodation rack **40** from opening portion **41c** (**S310**). Then, the CPU determines whether feeding section **91** has swung (**S320**). Specifically, when an OFF signal is input from the automatic switch **120a** and an ON signal is input from automatic switch **120b**, the CPU makes an affirmative determination. Meanwhile, when the OFF signal is input from automatic switches **120a** and **120b**, the CPU makes a negative determination.

[0109] This determination is performed for the following reasons. In a case where there is no support frame **100** in accommodation rack **40**, feeding section **91** (surface **93a** of belt **93**) does not abut on support frame **100** but misses. In this case, the piston of air cylinder **92** moves to the forward end, and automatic switch **120b** outputs the ON signal. In a case where there is support frame **100** in accommodation rack **40**, feeding section **91** (surface **93a** of belt **93**) abuts on support frame **100** but does not miss. In this case, the piston of air cylinder **92** is positioned between the backward end and the forward end, and automatic switches **120a** and **120b** output the OFF signal.

[0110] When an affirmative determination is made in **S320**, the CPU proceeds to **S360**. Meanwhile, when a negative determination is made in **S320**, the CPU controls drive section **94** such that support frame **100** is fed into accommodation rack **40** (**S330**). Next, the CPU waits until a

predetermined time elapses (S340). Here, the predetermined time is determined in advance as a time required for feeding out support frame **100**. Then, the CPU determines whether there is support frame **100** in the movement path (S350). The CPU acquires the detection signal of support frame detection sensor **24** from printing control section **28**. When the OFF signal is acquired, the CPU makes an affirmative determination. Meanwhile, when the ON signal is acquired, the CPU makes a negative determination. When an affirmative determination is made in S350, the CPU proceeds to S360.

[0111] After an affirmative determination is made in S320 or after an affirmative determination is made in S350, the CPU reports the error (S360). Specifically, the CPU outputs an error to management computer **130**. When an error is input, management computer **130** displays a message indicating that an error has occurred in a display device (not illustrated). When a negative determination is made in S350, the CPU controls air cylinder **92** such that feeding section **91** is pulled out from accommodation rack **40** through opening portion **41c** (S370)).

[0112] Next, the CPU returns to the member exchanging process routine, and controls lifting and lowering drive section **54** such that shelf section **41a** that accommodates support frame **100** that does not hold board support member **19** moves to a position facing exchange door **29** (S180). Next, the CPU executes a support frame feeding-out process subroutine (S190, refer to FIG. **18C**). When the support frame feeding-out process subroutine is started, the CPU controls air cylinder **92** such that feeding section **91** of delivery device **90** enters accommodation rack **40** from opening portion **41c** (S400).

[0113] Subsequently, the CPU determines whether feeding section **91** has swung (S410). When an affirmative determination is made in S410, the CPU proceeds to S460. Meanwhile, when a negative determination is made in S410, the CPU controls drive section **94** such that support frame **100** is fed out from accommodation rack **40** (S420).

[0114] Next, the CPU outputs a support frame feeding-in instruction to printing device **10** (S430). When the support frame feeding-in instruction is input, printing control section **28** performs a process of moving support frame **100** to mask work section **15** using exchange section **27**. Next, the CPU of exchange control section **110** waits until a predetermined time elapses (S440). Here, the predetermined time is determined in advance as a time required for feeding out support frame **100**. Subsequently, the CPU determines whether the movement path includes support frame **100** (S450). When an affirmative determination is made in S450, the CPU proceeds to S460. After an affirmative determination is made in S410 or after an affirmative determination is made in S450, the CPU reports the error (S460). When a negative determination is made in S450, the CPU controls air cylinder **92** such that feeding section **91** is pulled out from accommodation rack **40** through opening portion **41c** (S470).

[0115] Next, the CPU returns to the member exchanging process routine and outputs a collection instruction of board support member **19** to printing device **10** (S200). When the collection instruction is input, printing control section **28** controls a pusher (not illustrated) to push down claw member **103** of support frame **100**. Thus, claw member **103** is opened. In this state, printing control section **28** controls the board lifting and lowering section (not illustrated) to lift board support member **19**. Then, in this state, printing control section **28** releases the pressing of claw member **103** by the pusher. Accordingly, claw member **103** is closed, and board support member **19** is held by holding section **101** of support frame **100**.

[0116] Then, the CPU executes a support frame feeding-in process subroutine (S210). The support frame feeding-in process subroutine has already been described.

[0117] Subsequently, the CPU controls lifting and lowering drive section **54** such that shelf section **41a** accommodating support frame **100** holding board support member **19** which is the exchange target moves to a position facing exchange door **29** (S220). Next, the CPU executes the support frame feeding-out process subroutine (S230)). The support frame feeding-out process subroutine is as described above. Thus, support frame **100** holding board support member **19** which is the

exchange target is fed into printing device **10**.

[0118] Then, the CPU outputs an instruction to exchange board support member **19** to printing device **10** (**S240**). When the exchange instruction is input, printing control section **28** lifts the board lifting and lowering section (not illustrated). In this state, printing control section **28** controls the pusher (not illustrated) to push down claw member **103**. Thus, claw member **103** is opened, and board support member **19** is transferred to the board lifting and lowering section.

[0119] Next, the CPU executes the support frame feeding-in process subroutine (**S250**). Then, the CPU of exchange control section **110** controls lifting and lowering drive section **54** such that shelf section **41a** that accommodates support frame **100** holding screen mask M moves to a position facing exchange door **29** (**S260**).

[0120] Next, the CPU executes the support frame feeding-out process subroutine (**S270**), and controls lifting and lowering drive section **54** such that accommodation rack **40** moves to the operation position (**S280**).

[0121] Subsequently, the CPU releases the clamp by fixing section **70** and moves the hook section to the accommodation position (**S290**). After **S360** of the support frame feeding-in process subroutine, after **S460** of the support frame feeding-out process subroutine or after **S290** of the member exchanging process routine, the CPU ends this routine.

[0122] Here, a correspondence relationship between the elements of the present embodiment and the elements of the present disclosure will be clarified. Printing system **2** of the present embodiment corresponds to a printing system of the present disclosure, printing device **10** corresponds to a printing device, accommodation rack **40** corresponds to an accommodation rack, delivery device **90** corresponds to a delivery device, position sensor **120** and exchange control section **110** correspond to a first detection section, and support frame detection sensor **24** and printing control section **28** correspond to a second detection section.

[0123] In printing system **2** described above, it is possible to detect that there is no support frame **100** in shelf section **41a** based on the detection signal of position sensor **120**. Therefore, it is possible to prevent support frame **100** from being fed out to printing device **10** in a state where there is no support frame in shelf section **41a**, or to prevent support frame **100** from being fed in from printing device **10** in a state where there is the support frame in shelf section **41a**. As a result, support frame **100** can be more reliably fed in or out between accommodation rack **40** and printing device **10**.

[0124] Further, printing system **2** includes support frame detection sensor **24** capable of detecting the presence or absence of an object in the movement path between shelf section **41a** and printing device **10**, and includes exchange control section **110** capable of detecting that support frame **100** is caught in the movement path based on the detection signal from support frame detection sensor **24**. Therefore, it is possible to prevent a situation in which accommodation rack **40** is lifted and lowered in a state where support frame **100** is caught in the movement path.

[0125] Further, in printing system **2**, weight **49** is provided in accommodation rack **40** such that the center of gravity of accommodation rack **40** is positioned closer to printing device **10** side than the position supported by rack support member **53**. Therefore, when accommodation rack **40** is lifted and lowered together with rack support member **53**, accommodation rack **40** is less likely to be inclined to the side opposite to printing device **10**. Further, the present disclosure is particularly effective in printing system **2** since holding section **101** on the upstream side in the movement direction of support frame **100** holds board support member **19** having a larger size (heavy) than board support member **19** held by holding section **101** on the downstream side in the movement direction of support frame **100**. In this case, compared to a case where board support member **19** having the same size is held in the holding sections **101** on the downstream side and the upstream side, or in a case where board support member **19** is held only on the holding section on the downstream side in the movement direction, the center of gravity of accommodation rack **40** is likely to be biased to the side opposite to printing device **10** with respect to rack support member

53.

[0126] The present disclosure is not limited to the embodiment described above at all, and needless to say, the present disclosure can be carried out in various forms as long as the forms belong to the technical scope of the present disclosure.

[0127] In the above-described embodiment, delivery device **90** is used to feed out or in support frame **100** between printing device **10** and accommodation rack **40**. However, instead of delivery device **90**, delivery device **290** illustrated in FIGS. **24** and **25** may be used. Delivery device **290** includes feeding section **191** and air cylinder **92**. Feeding section **191** includes drive section **94**, belt **293**, and multiple (two in FIG. **24**) driving rolls **296**. Belt **293** transmits the power of drive section **94** to driving roll **296** via multiple pulleys **95**. Belt **293** is made of rubber, for example. Driving roll **296** is capable of feeding out or in support frame **100** in the movement direction (the front-rear direction) by being driven in a state of abutting on support frame **100**. Driving roll **296** is formed of a material (for example, urethane) having a lower abrasiveness than belt **293**. This makes it difficult for the abrasion powder to enter the inside of the printing device as compared with the case where the support frame is fed out or in by the friction force of the belt. This is because the urethane roll is less likely to wear than the belt.

[0128] In the above-described embodiment, exchange unit **30** includes hook section **56** and engages with block section **45** of accommodation rack **40**. However, a mechanism other than hook section **56** may be connected to accommodation rack **40**. In addition, in the embodiment that has been described heretofore, exchange control section **110** is provided for causing hook section **56** to be lifted and lowered, however, this exchange control section may be omitted.

[0129] In the embodiment that has been described heretofore, exchange unit **30** is described as including clamp section **71** and stopper section **75** as fixing section **70**, however, the present disclosure is not particularly limited thereto, and hence, either of them may be omitted. In addition, any other fixing section than clamp section **71** and stopper section **75** may be adopted.

Alternatively, although exchange unit **30** is described as including fixing section **70**, this fixing section **70** may be omitted.

[0130] In the above-described embodiment, automatic switches **120a** and **120b** are provided. However, only automatic switch **120b** may be provided.

[0131] In the embodiment that has been described heretofore, worker **W** is described as moving accommodation rack **40**, however, the present disclosure is not particularly limited thereto, and hence, for example, accommodation rack **40** may be moved by an automatic conveyance vehicle. The automatic conveyance vehicle may be configured to move accommodation rack **40** by entering the space in the center of caster section **43**. Here, handle **42** may be omitted in the case that automatic conveyance vehicle moves accommodation rack **40**.

[0132] In the embodiment described above, exchange section **27** is described as being made into the rod structure installed in printing head **12** and having the abutting section, however, the present disclosure is not particularly limited thereto, and hence exchange section **27** may be made into an independent mechanism installed in parts other than printing head **12**. In addition, in the embodiment described above, exchange section **27** is described as being made into the rod mechanism configured to be brought into engagement with support frame **100**, however, exchange section **27** may be made into, for example, a conveyor mechanism on which support frame **100** including an exchangeable member is placed for movement.

[0133] In the embodiment described heretofore, accommodation rack **40** is described as having no drive section that moves support frame **100**, however, the present disclosure is not particularly limited thereto, and hence, accommodation rack **40** may be configured as one having a drive section that moves support frame **100**.

[0134] In the above-described embodiment, delivery device **90** is provided only in left unit **51**. However, delivery device **90** may be provided only in right unit **61**, or may be provided in both of left unit **51** and right unit **61**.

[0135] In the embodiment described above, the exchangeable member is described as being screen mask M or board support member 19, however, the present disclosure is not limited thereto as long as the exchangeable member is a member needing to be exchanged, and hence the exchangeable member may be one or more of parts, in addition to screen mask M or board support member 19, including cartridge 23 for accommodating the viscous fluid, squeegee 14, cleaning member 20a for use in cleaning screen mask M, and the like. In addition, the processing target is described as being board S; however, the present disclosure is not particularly limited thereto, as long as the processing target constitutes an object on which viscous fluid can be printed, and hence, the processing target may be a three-dimensional object or the like. In addition, although the viscous fluid is described as being the solder paste, conductive paste, an adhesive, or the like may be used.

Industrial Applicability

[0136] The present disclosure is applicable to a technical field of devices for performing a mounting process of a component.

REFERENCE SIGNS LIST

[0137] 1: mounting system, 2: printing system, 3: print inspection device, 4: mounting device, 5: mounting inspection device, 10: printing device, 11: printing section, 12: printing head, 13: printing moving section, 14: squeegee, 15: mask work section, 17: board processing section, 18: board conveyance section, 19: board support member, 20: cleaning section, 20a: cleaning member, 21: supply section, 23: cartridge, 24: support frame detection sensor, 24a: light projecting section, 24b: light receiving section, 25: rack detection sensor, 26: operation panel, 27: exchange section, 28: printing control section, 29: exchange door, 30: exchange unit, 40: accommodation rack, 41: rack main body, 41a: shelf section, 41b: passage section, 41c: opening portion, 42: handle, 43: caster section, 44: supporting and fixing member, 45: block section, 46: receiving section, 47: locking hole, 48: insertion hole, 49: weight, 51: left unit, 52: lifting and lowering device, 53: rack support member, 53a: cam follower, 54: lifting and lowering drive section, 55: movement device, 56: hook section, 57: drive section, 58: lifting and lowering space, 61: right unit, 70: fixing section, 71: clamp section, 72: clamp lever, 73: lever rotating section, 74: air cylinder, 75: stopper section, 76: insertion pin, 77: pin drive section, 78: spring, 79: cam follower, 80: guide member, 81: air cylinder, 90: delivery device, 91: feeding section, 92: air cylinder, 93: belt, 93a: surface, 94: drive section, 95: pulley, 100: support frame, 101: holding section, 103: claw member, 110: exchange control section, 120: position sensor, 120a, 120b: automatic switch, 130: management computer, 191: feeding section, 290: delivery device, 293: belt, 296: driving roll, L: cross section, M: screen mask, S: board, W: worker

Claims

1. A printing system comprising: a printing device; an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; a delivery device including a feeding section capable of feeding in or out the support frame in the movement direction in a state of abutting on the support frame in the accommodation rack, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening; and a first detection section including a first sensor configured to detect an entering state of the feeding section with respect to the accommodation rack, and capable of detecting that there is no support frame in the shelf sections based on a detection signal of the first sensor.

2. The printing system according to claim 1, further comprising: a second detection section including a second detection sensor capable of detecting presence or absence of an object on a movement path between the shelf section and the printing device, and capable of detecting that the

support frame is caught on the movement path based on a detection signal from the second sensor.

3. A printing system comprising: a printing device; an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device; and a lifting and lowering device including a pair of rack support members capable of supporting a lower surface of the accommodation rack from both sides in an orthogonal direction orthogonal to a movement direction of the support frame and an up-down direction in the state where the accommodation rack faces the printing device, and configured to lift and lower the rack support members, wherein the support frame includes multiple holding sections arranged in the movement direction, the holding sections being capable of holding exchanging members of different sizes, and the accommodation rack includes a weight provided such that a center of gravity of the accommodation rack is positioned closer to the printing device than a position supported by the rack support members in the state where the accommodation rack faces the printing device.

4. A printing system comprising: a printing device; an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; and a delivery device including a feeding section including one or more driving rolls made of urethane, and capable of feeding in or out the support frame in the movement direction by the driving rolls being driven in a state where the driving rolls abut on the support frame in the accommodation rack, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening.

5. A printing system comprising: a printing device; an accommodation rack in which multiple shelf sections capable of accommodating a support frame supporting an exchanging member are provided and the support frame is movable between the printing device and the shelf sections in a state where the accommodation rack faces the printing device, and which has an opening in a side surface parallel to a movement direction of the support frame; and a delivery device including a feeding section capable of feeding in or out the support frame in the movement direction, and a moving section causing the feeding section to enter and exit from the accommodation rack through the opening, wherein the feeding section includes one or more driving rolls driven in a state of abutting on the support frame in the accommodation rack, and a belt provided at a position not in contact with the support frame to transmit power to the driving rolls, and the driving rolls have a lower abrasiveness than the belt.
